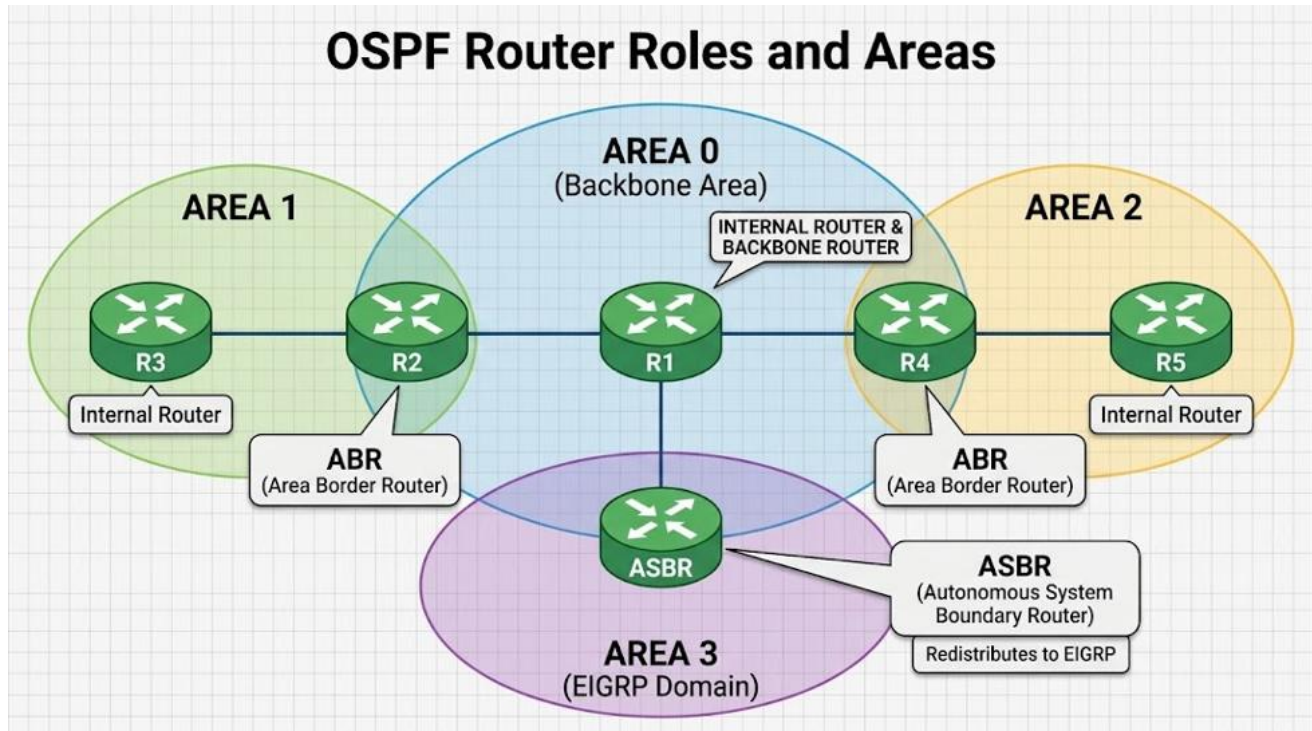


OSPF MULTI-AREA & MULTIPLE LSA

LAB WORKBOOK

CCNA / Cisco IOS Hands-on Practice



Focus: LSA Type 1, 2, 3, 4, 5 and 7 • Internal Router • ABR • ASBR • NSSA

1. Learning Objectives

- Build OSPF Area 0, Area 1 and Area 2 using Cisco IOS routers.
- Identify Internal Routers, Area Border Routers (ABRs) and Autonomous System Boundary Routers (ASBRs).
- Observe LSA Type 1 (Router), Type 2 (Network), Type 3 (Summary), Type 4 (ASBR Summary), Type 5 (External) and Type 7 (NSSA External).
- Use show ip ospf database commands to locate LSAs and understand what each LSA represents.
- Configure route summarization and verify how Type 3 LSAs change.
- Redistribute an external route and verify Type 4 and Type 5 LSAs.
- Build an NSSA and observe Type 7-to-Type 5 translation at an ABR.
- Troubleshoot common multi-area OSPF problems using show and debug commands.

2. LSA Quick Reference

LSA	Name	Generated by	Main purpose	Where seen
Type 1	Router LSA	Every OSPF router	Describes router links/interfaces and OSPF topology inside an area.	Only inside its originating area
Type 2	Network LSA	DR on multi-access network	Describes routers attached to a shared multi-access segment.	Only inside its originating area
Type 3	Summary LSA	ABR	Advertises networks from one area into another area.	Other OSPF areas
Type 4	ASBR Summary LSA	ABR	Provides reachability to an ASBR so routers can reach external routes.	Other areas
Type 5	AS External LSA	ASBR	Advertises routes redistributed into OSPF.	OSPF domain except stub areas
Type 7	NSSA External LSA	ASBR inside NSSA	Carries redistributed external routes inside an NSSA.	Only inside NSSA; ABR can translate to Type 5

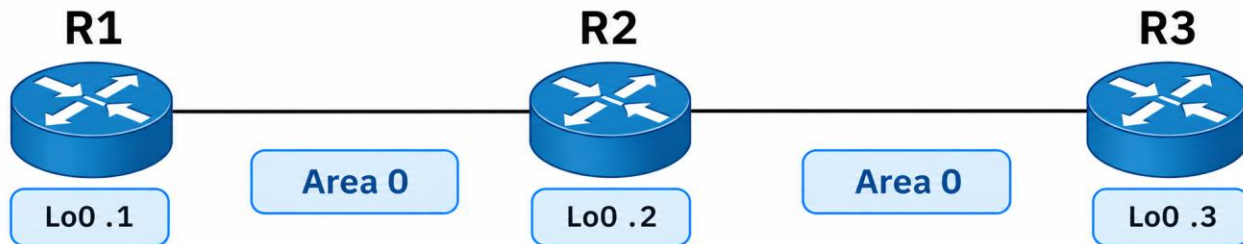
3. Recommended Addressing Plan

Link / LAN	Network	Device / Interface	IP
R1-R2	10.0.12.0/30	R1 G0/0	10.0.12.1
R1-R2	10.0.12.0/30	R2 G0/0	10.0.12.2
R2-R3	10.0.23.0/30	R2 G0/1	10.0.23.1
R2-R3	10.0.23.0/30	R3 G0/0	10.0.23.2
R1-R4	10.0.14.0/30	R1 G0/1	10.0.14.1
R1-R4	10.0.14.0/30	R4 G0/0	10.0.14.2
R4-R5	10.0.45.0/30	R4 G0/1	10.0.45.1
R4-R5	10.0.45.0/30	R5 G0/0	10.0.45.2
R1 Loopback	192.168.10.0/24	R1 Lo0	192.168.10.1
R3 Loopback	192.168.30.0/24	R3 Lo0	192.168.30.1
R5 Loopback	192.168.50.0/24	R5 Lo0	192.168.50.1

LAB 1 — OSPF Single Area: Observe LSA Type 1

Goal: Build a basic Area 0 and verify Router LSAs. This is the foundation for all later labs.

Topology



Tasks

1. Configure IP addresses on the three routers.
2. Enable OSPF process 1 and place all interfaces in Area 0.
3. Use router IDs 1.1.1.1, 2.2.2.2 and 3.3.3.3.
4. Verify OSPF neighbors.
5. Inspect the Router LSDB and identify one Type 1 LSA per router.

R1 Configuration

```
hostname R1
interface g0/0
  ip address 10.0.12.1 255.255.255.252
  no shutdown
interface loopback0
  ip address 192.168.10.1 255.255.255.0

router ospf 1
  router-id 1.1.1.1
  network 10.0.12.0 0.0.0.3 area 0
  network 192.168.10.0 0.0.0.255 area 0
```

R2 Configuration

```
hostname R2
interface g0/0
  ip address 10.0.12.2 255.255.255.252
  no shutdown
interface g0/1
  ip address 10.0.23.1 255.255.255.252
  no shutdown
interface loopback0
  ip address 192.168.20.1 255.255.255.0

router ospf 1
  router-id 2.2.2.2
  network 10.0.12.0 0.0.0.3 area 0
  network 10.0.23.0 0.0.0.3 area 0
```

```
network 192.168.20.0 0.0.0.255 area 0
```

R3 Configuration

```
hostname R3
interface g0/0
 ip address 10.0.23.2 255.255.255.252
 no shutdown
interface loopback0
 ip address 192.168.30.1 255.255.255.0

router ospf 1
 router-id 3.3.3.3
 network 10.0.23.0 0.0.0.3 area 0
 network 192.168.30.0 0.0.0.255 area 0
```

Verification

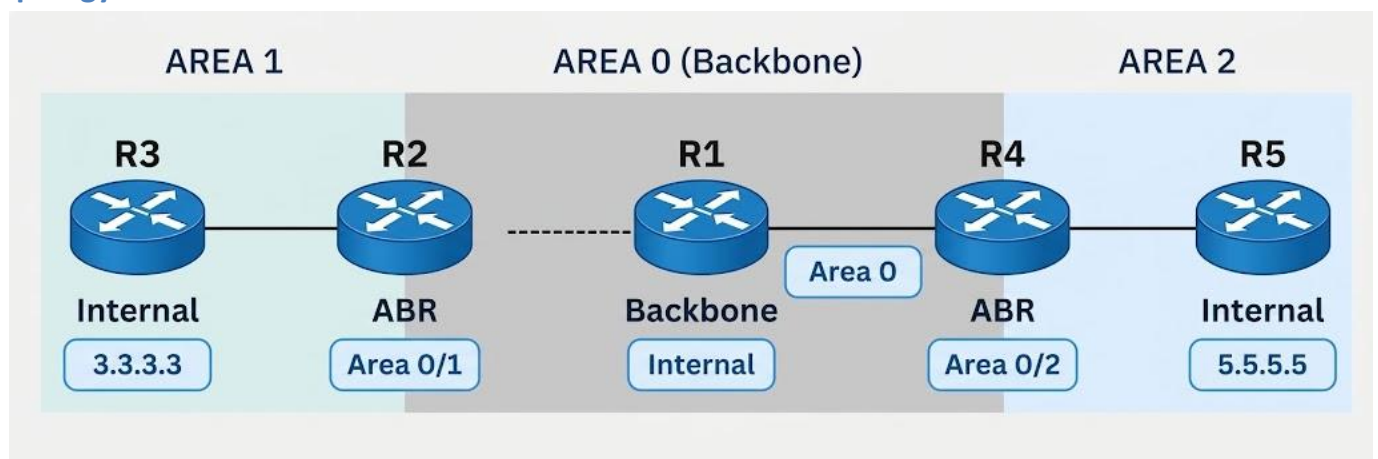
```
show ip ospf neighbor
show ip route ospf
show ip ospf database
show ip ospf database router
show ip ospf database router 1.1.1.1
```

Expected result: Each router originates a Type 1 Router LSA. Because all routers are in Area 0, the Type 1 LSAs remain inside Area 0.

LAB 2 — Multi-Area OSPF: LSA Type 1 + Type 3

Goal: Create Area 1 and Area 2 around the Area 0 backbone. R2 and R4 become ABRs.

Topology



Area Assignment

Router	Interface	Area	Role
R3	G0/0	1	Internal Router
R2	G0/1	1	ABR
R2	G0/0	0	ABR
R1	G0/0/G0/1	0	Internal Backbone Router
R4	G0/0	0	ABR

R4	G0/1	2	ABR
R5	G0/0	2	Internal Router

R2 ABR Configuration

```
hostname R2
interface g0/0
 ip address 10.0.12.2 255.255.255.252
 no shutdown
interface g0/1
 ip address 10.0.23.1 255.255.255.252
 no shutdown
interface loopback0
 ip address 192.168.20.1 255.255.255.0

router ospf 1
 router-id 2.2.2.2
 network 10.0.12.0 0.0.0.3 area 0
 network 10.0.23.0 0.0.0.3 area 1
 network 192.168.20.0 0.0.0.255 area 0
```

R3 Area 1 Configuration

```
router ospf 1
 router-id 3.3.3.3
 network 10.0.23.0 0.0.0.3 area 1
 network 192.168.30.0 0.0.0.255 area 1
```

R4 ABR Configuration

```
hostname R4
interface g0/0
 ip address 10.0.14.2 255.255.255.252
 no shutdown
interface g0/1
 ip address 10.0.45.1 255.255.255.252
 no shutdown
interface loopback0
 ip address 192.168.40.1 255.255.255.0

router ospf 1
 router-id 4.4.4.4
 network 10.0.14.0 0.0.0.3 area 0
 network 10.0.45.0 0.0.0.3 area 2
 network 192.168.40.0 0.0.0.255 area 0
```

R5 Area 2 Configuration

```
router ospf 1
 router-id 5.5.5.5
 network 10.0.45.0 0.0.0.3 area 2
 network 192.168.50.0 0.0.0.255 area 2
```

Verification — Find Type 3

```
show ip ospf neighbor
show ip ospf database
show ip ospf database summary
show ip ospf database summary 192.168.30.0
```

```
show ip route ospf
```

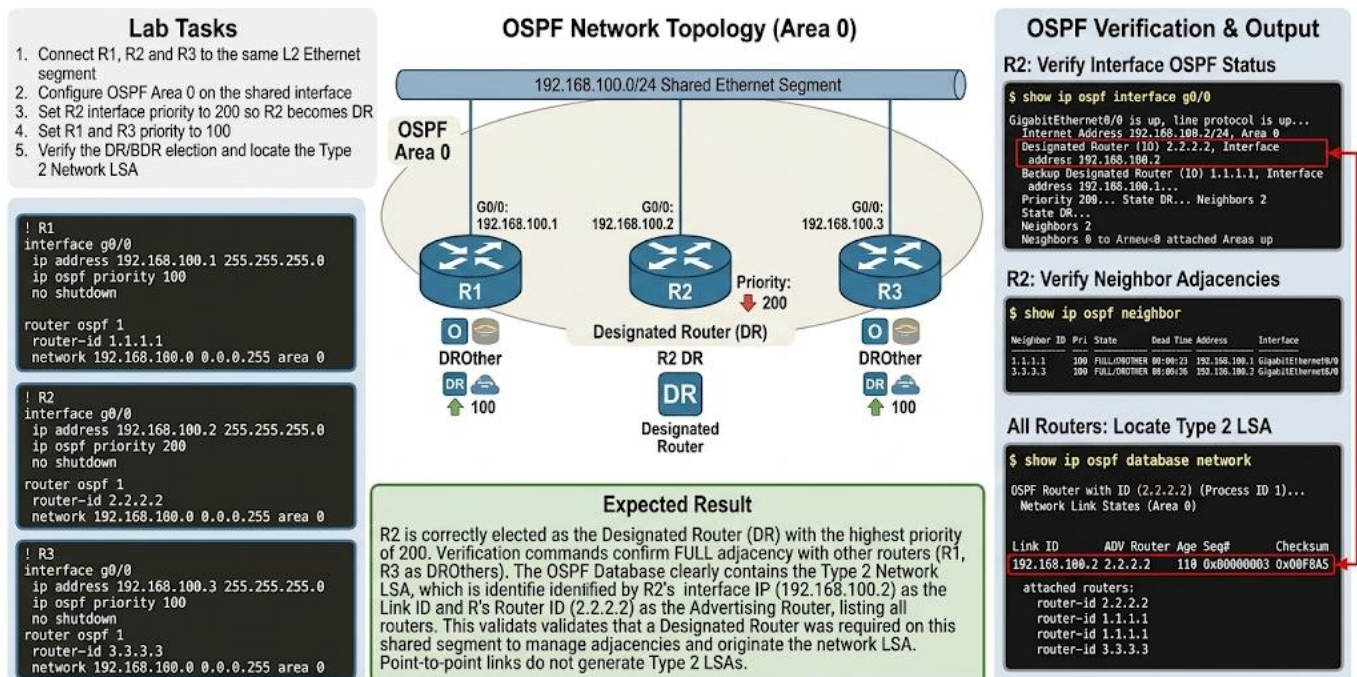
Expected result: R2 advertises Area 1 networks toward Area 0 as Type 3 Summary LSAs. R4 advertises Area 2 networks toward Area 0, and the ABRs advertise inter-area routes into the other areas.

LAB 3 — OSPF Broadcast Network: LSA Type 2

Goal: Observe the Type 2 Network LSA generated by a DR on a multi-access Ethernet segment.

Topology

OSPF Lab: Designated Router (DR) Election & Type 2 LSA



Tasks

6. Connect R1, R2 and R3 to the same Layer-2 Ethernet segment.
7. Configure OSPF Area 0 on the shared interface.
8. Set R2 interface priority to 200 so R2 becomes DR.
9. Set R1 and R3 priority to 100.
10. Verify the DR/BDR election and locate the Type 2 Network LSA.

```
! R2
interface g0/0
ip address 192.168.100.2 255.255.255.0
ip ospf priority 200
no shutdown
router ospf 1
router-id 2.2.2.2
network 192.168.100.0 0.0.0.255 area 0
```

```
! R1
interface g0/0
```

```

ip address 192.168.100.1 255.255.255.0
ip ospf priority 100
no shutdown
router ospf 1
router-id 1.1.1.1
network 192.168.100.0 0.0.0.255 area 0

! R3
interface g0/0
ip address 192.168.100.3 255.255.255.0
ip ospf priority 100
no shutdown
router ospf 1
router-id 3.3.3.3
network 192.168.100.0 0.0.0.255 area 0

show ip ospf interface g0/0
show ip ospf neighbor
show ip ospf database network

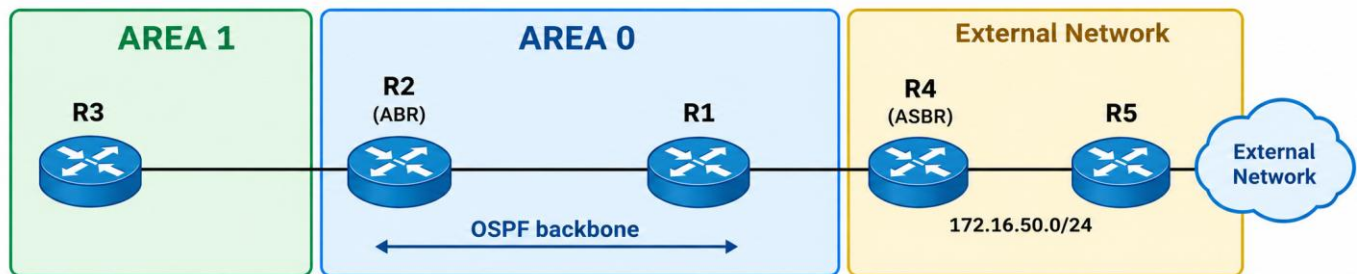
```

Expected result: R2 is the DR. The Type 2 Network LSA is identified by the DR's router ID and the shared network's IP address. Type 2 LSAs are not generated on point-to-point links.

LAB 4 — Route Redistribution: LSA Type 4 + Type 5

Goal: Introduce an external route through an ASBR and identify the Type 5 external LSA and the Type 4 ASBR Summary LSA.

Topology



R4 is an ASBR because it redistributes a non-OSPF route into OSPF.

R4 ASBR Configuration

```

interface g0/0
ip address 10.0.14.2 255.255.255.252
no shutdown
interface loopback50
ip address 172.16.50.1 255.255.255.0

router ospf 1
router-id 4.4.4.4
network 10.0.14.0 0.0.0.3 area 0

```

```
redistribute connected subnets
```

For a cleaner demonstration, you may instead use a static route and redistribute static:

```
ip route 172.16.60.0 255.255.255.0 192.168.50.2
```

```
router ospf 1
redistribute static subnets
```

Verification

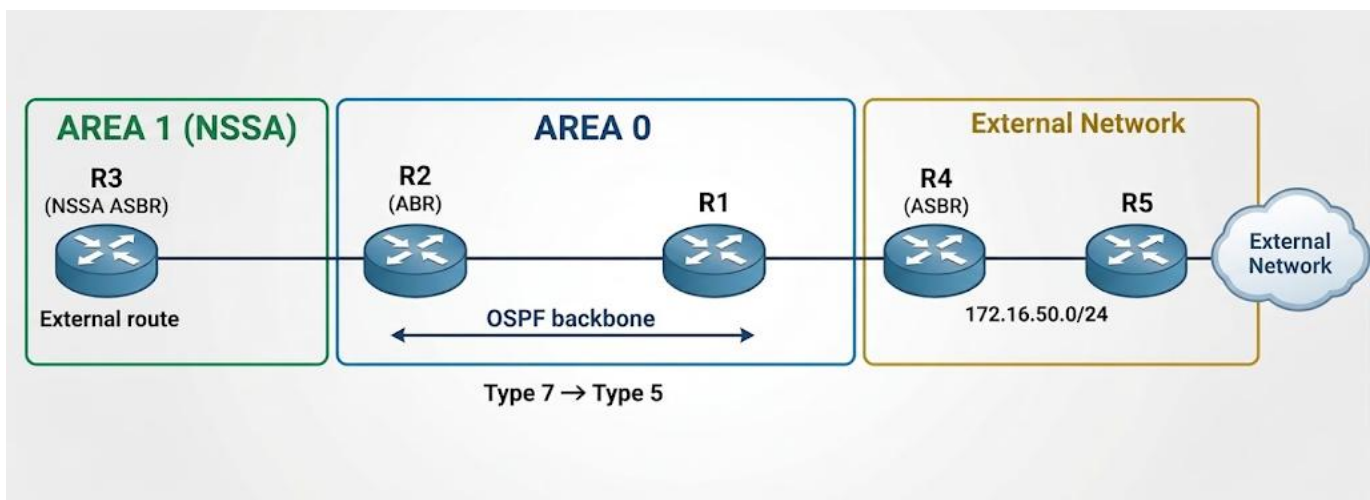
```
show ip ospf database external
show ip ospf database asbr-summary
show ip route ospf
show ip route 172.16.50.0
```

Expected result: R4 originates a Type 5 AS External LSA for the redistributed network. An ABR in another area can originate a Type 4 ASBR Summary LSA describing how to reach R4, the ASBR. Type 4 identifies the ASBR; Type 5 carries the external destination.

LAB 5 — NSSA: LSA Type 7 and Type 7-to-Type 5 Translation

Goal: Create a Not-So-Stubby Area (NSSA) and observe a Type 7 external LSA generated inside the NSSA. The ABR translates the Type 7 into a Type 5 for the rest of the OSPF domain.

Topology



R2 ABR

```
interface g0/0
ip address 10.0.12.2 255.255.255.252
no shutdown
interface g0/1
ip address 10.0.23.1 255.255.255.252
no shutdown

router ospf 1
router-id 2.2.2.2
network 10.0.12.0 0.0.0.3 area 0
network 10.0.23.0 0.0.0.3 area 1
```

```
area 1 nssa
```

R3 NSSA ASBR

```
interface g0/0
 ip address 10.0.23.2 255.255.255.252
 no shutdown
interface loopback70
 ip address 172.16.70.1 255.255.255.0

router ospf 1
 router-id 3.3.3.3
 network 10.0.23.0 0.0.0.3 area 1
 redistribute connected subnets
 area 1 nssa
```

Verification

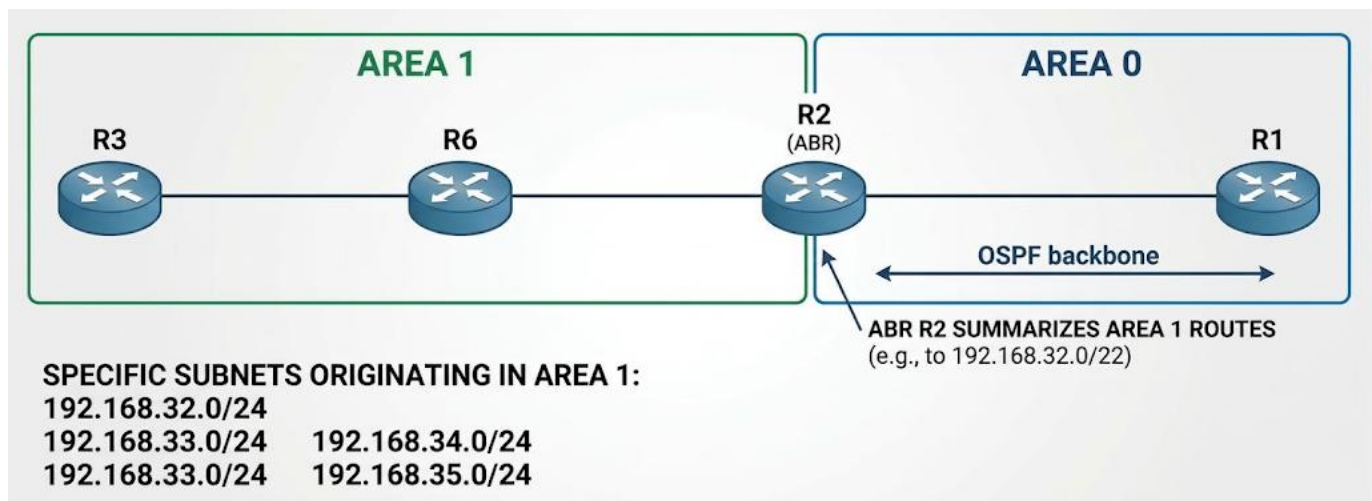
```
show ip ospf
show ip ospf database nssa-external
show ip ospf database external
show ip route ospf
```

Expected result: R3 originates a Type 7 LSA inside Area 1. R2, as the ABR, translates eligible Type 7 information into a Type 5 LSA toward Area 0. Use the LSDB to compare the Type 7 and Type 5 entries.

LAB 6 — Route Summarization: Control Type 3 LSAs

Goal: Summarize multiple Area 1 routes at the ABR and observe how the Type 3 LSDB changes.

Topology



Example Summary

The four /24 networks can be summarized as 192.168.32.0/22 if the addressing is contiguous and the summary does not include unwanted networks.

```
router ospf 1
 area 1 range 192.168.32.0 255.255.252.0
```

Verification

```
show ip ospf database summary
show ip route ospf
show ip ospf database summary 192.168.32.0
```

Expected result: The ABR can advertise the summary as a Type 3 LSA rather than advertising every individual Area 1 prefix toward Area 0.

Multi-LSA Troubleshooting Challenge

Problem	Symptom	Useful commands	Expected investigation
Wrong OSPF area	Neighbor does not form	show ip ospf interface; show ip ospf neighbor	Both ends of a link must agree on the OSPF area.
Duplicate router ID	Unexpected neighbor/LSDB behavior	show ip ospf; show ip ospf neighbor	Router IDs must be unique in the OSPF domain.
Interface passive	No adjacency on the link	show ip protocols; show run sec router ospf	Remove passive-interface from the transit link.
Missing network statement	Connected network absent from LSDB	show ip ospf interface brief; show ip ospf database	Place the correct interface/network in OSPF.
Area 1 not NSSA on ABR	NSSA adjacency/problem	show ip ospf; show run sec router ospf	Both sides of the area must use matching area type.
Redistribution missing	No external route	show ip route; show ip ospf database external	Verify redistribution and source route.
Unexpected DR	Different DR than expected	show ip ospf neighbor; show ip ospf interface	Check priority and restart/re-elect adjacency if required.

8. OSPF LSA Verification Cheat Sheet

```
show ip ospf
show ip ospf neighbor
show ip ospf interface brief
show ip ospf interface
show ip route ospf
```

```
show ip ospf database
show ip ospf database router
show ip ospf database network
show ip ospf database summary
show ip ospf database asbr-summary
show ip ospf database external
show ip ospf database nssa-external
```

```
show ip ospf database router self-originate
show ip ospf database summary self-originate
show ip ospf database external self-originate
```

10. Practical Questions

11. Why does a Type 1 LSA not cross an OSPF area boundary?
12. Why is a Type 2 LSA associated with a DR?
13. What is the difference between a Type 3 LSA and a Type 5 LSA?
14. What information does a Type 4 LSA provide?
15. What is the role of an ABR?
16. What makes a router an ASBR?
17. Why is Type 7 used in an NSSA?
18. Where does Type 7 get translated into Type 5?
19. How can you identify the advertising router for an LSA?
20. Which show command would you use first when an OSPF neighbor is missing?